

Toward automated construction: The design-to-printing workflow for a robotic in-situ 3D printed house

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ABSTRACT

In the construction industry, the labor shortage is a global issue. Characterized by the forming process of layered extrusion and rapid solidification, 3D concrete printing (3DCP) technology begins to be used in digital concrete construction to minimize the demand for manpower. However, the utilization of 3DCP in full-scale architectural constructions is still in its infancy, involving many manual processes like rebars-binding and formwork-making, which reduces the benefits of construction automation that should be brought by such technology. Therefore, this study proposes a design-to-printing workflow to 3D print the entire structure of a farmhouse in Wujiazhuang (WJZ) village. Firstly, design strategies are built up by a series of multi-disciplinary research, including printing system development, site planning, material design, and structure evaluation. Secondly, construction strategies involving printing process organization and man-hour-saving construction methods are utilized in the 3D printing process. By applying the introduced workflow, the project saves 62.4 % of human resources compared with the traditional reinforced concrete (RC) method, and the participation of 3D printing technologies arises from 18.3 % in the common 3DCP scenario to 76.5 % in this project. The on-site construction duration of the WJZ farmhouse is also reduced by 24.5 % in comparison with the unplanned construction process.

1. Background

The construction industry is facing the problem of labor shortage [1]. According to Associated Builders and Contractors, the construction industry in the United States faces a workforce shortage of 650,000 in 2022 [2]. A report released by the National Bureau of Statistics of China also revealed that the total number of migrant workers in China in 2020 decreased by 5,170,000 compared with the previous year [3]. Besides, problems like the declining productivity in the construction industry [4,5] and the occupational health risks brought by workplace environmental conditions [6] all force the construction industry to transform and upgrade.

Correspondingly, it is a remarkable transformation that 3D printing technology has been changing the design and manufacturing

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processes of the construction industry [7]. Compared with cast-in-situ reinforced concrete and precast concrete, 3D concrete printing technology replaces the typical "molding-solidifying" process with layered extrusion, hence eliminating the reliance on formwork, simplifying manual steps, and reducing the manpower [8]. Furthermore, the numerical-controlled printing process of 3DCP takes the place of some risky operations for humans and improves the safety of the worksite [9].

However, there are still challenges in applying 3DCP to real-scale building constructions. Limited by material properties and the mechanism of printers, extra processes that consume manpower, time, and materials will inevitably appear in the concrete construction process, like manually inserting reinforcement ties in the printing process [10] or pre-binding the reinforcement cages of columns [11]. In short, although the transition from formwork-based manufacturing to 3D printing is claimed to trigger a paradigm shift in concrete technology [12], such a transition hasn't yet happened on the building construction site. In each step of architecture-scale 3D concrete printing, the manufacturing procedures still have plenty of room to optimize. Therefore, this paper will introduce a well-organized design-to-printing workflow of a concrete farmhouse, which could provide a reference for future in-situ 3DCP building projects.

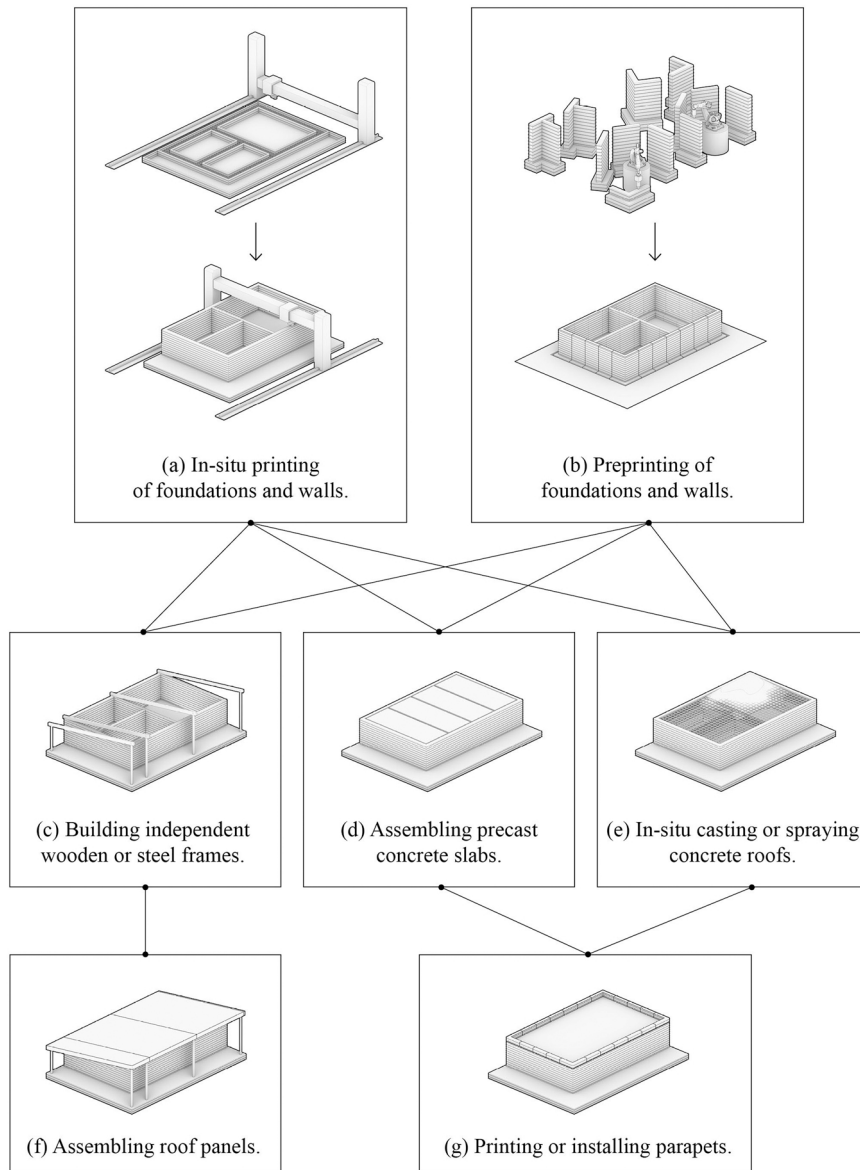


Fig. 1. Typical 3DCP building construction processes.

2. Introduction

2.1. Literature review

Collectively, the selections of materials, printers, and construction methods contribute to the realization of a 3DCP building. Therefore, the literature review will introduce current research and practices in the construction industry from the above three aspects.

2.1.1. 3D printing materials in architecture-scale

The utilization of 3D printing materials and technologies varies in different industrial domains. For instance, glass is increasingly desired for digital light manufacturing of complex products like micro-optics and microfluidic systems [13], and polyamide powders are adopted in the plasma process method to develop technical and decorative coatings [14,15]. In the construction industry, Portland-cement-based concrete is the most commonly used material. However, the production process of Portland cement (PC) is highly energy intensive and results in high carbon footprints [16,17]. In this regard, alternative environmental materials, such as geopolymers [16–18] and sulpho-aluminate cement (SC), are introduced into the industry to achieve fewer carbon emissions in the 3DCP building life cycle. Specifically, it is proved that sulpho-aluminate cement not only releases much less carbon dioxide than Portland cement in the material production process but also shows printable properties like high early-age strengths and short setting times [19,20,52]. Therefore, the development of sulpho-aluminate cementitious materials is of great potential in architecture-scale 3D printing.

Additionally, sulpho-aluminate cement is proved to have better adaptability in low-temperature and low-humidity environments compared with Portland cement. Firstly, even though the hydrations of both SC and PC proceed slowly at low temperature ($< 5^{\circ}\text{C}$) and occur rapidly at high temperature ($> 40^{\circ}\text{C}$) [52,53], SC has the advantage of favorable anti-freeze properties compared with PC, it can even be used as a mineral accelerator for PC at low temperatures [54]. Furthermore, SC displays relatively low drying shrinkage than PC. Namely, in low-humidity environments, the deformation caused by the evaporation of water inside SC is smaller than that inside PC [55,56].

2.1.2. 3D printers and construction environments

Currently, the robotic arm and the gantry printer are the most frequently used 3D concrete printers, and they are suitable for different construction environments. Benefiting from the small size and six-axis moving mechanism, the robotic arm can suit compacted built environments and realize detailed and complex building forms [21]. However, since the printable area of a single robotic arm is relatively small, most robotic 3DCP projects are preprinted in the factory rather than printed in-situ, which fails to give full play to the flexibility of robotic arms to suit compacted construction sites.

On the contrary, the gantry printer owns a larger size and simple-operated 3-axis movement, which makes it desirable for printing regular building shapes in open construction environments. Since the structure of the gantry girder limits its printhead inside the frame, so the gantry printer might not be used to finish a project located in high-density environments (e.g. crowded villages). Considering that the printing area of the WJZ farmhouse is located within a building complex, we decided to adopt robotic arms in in-

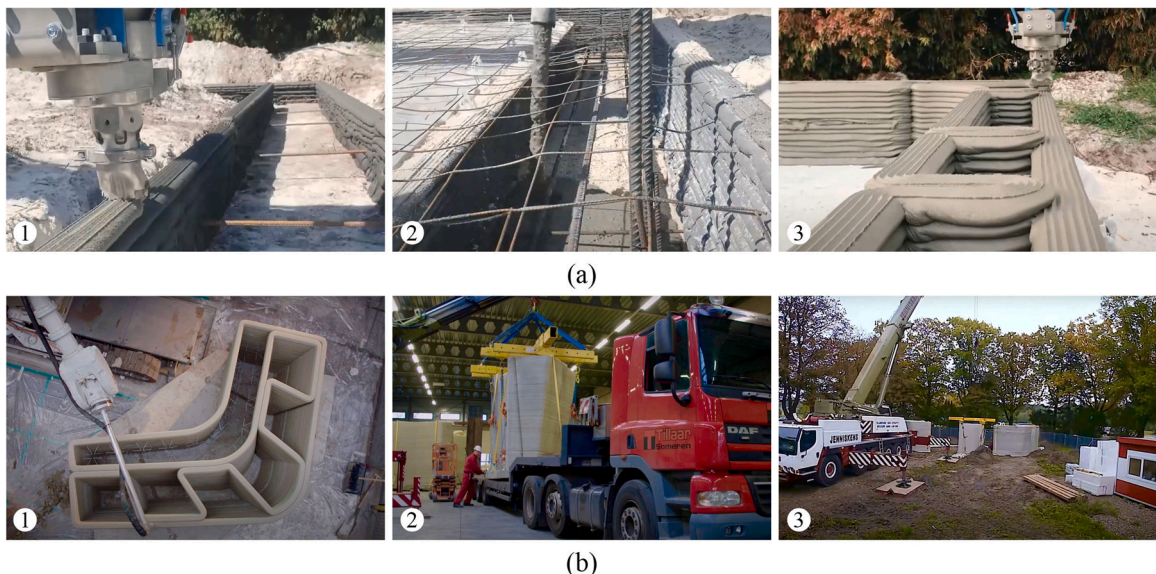


Fig. 2. (a) Construction records of the Florida Storage Unit project [22]: (1) In-situ printing and reinforcing of the trenches; (2) Casting the reinforced trenches; (3) In-situ printing of the walls. (b) Construction records of the Project Milestone [24]: (1) Pre-printing the foundation-to-wall component; (2) Transportation process of preprinted components; (3) Assembling process of the house.

situ construction and develop their abilities in mobility.

2.1.3. 3DCP workflow in real construction industry

In an architectural scale 3DCP project, 3D printing technology still has to follow the formal and functional requirements of various structural components, including foundations, walls, and roofs. Therefore, different from printing furniture or facilities, the 3D printing process of buildings often needs to be divided into multiple phases with specific printing methods. By researching and categorizing the current 3DCP projects of real-scale architectures (Fig. 1), the workflow of 3DCP can be divided into two main phases. The former is the foundations and walls construction that is based on the 3D printing technology and the latter is the roof construction that is usually realized by prefabrication or in-situ casting.

The methods to print foundations and walls mainly include in-situ printing and preprinting. In in-situ printing (Fig. 1a), the actual 3D printed part of foundations are the trenches, where rebars are manually embedded for later casting. Then walls can be printed in their actual positions after the cast foundations are solidified. Besides, the bottoms of walls should also be poured with mortar in the printing process so that walls and foundations can be connected by embedded rebars. Subsequently, all building walls can be printed continuously in one go. A typical example of this method is the Florida Storage Unit printed by the COBOD gantry printer (Fig. 2a) [22]. In preprinting (Fig. 1b), the building model will be divided into several transportable components and each component integrates parts of foundations and walls, like the Project Milestone from the Eindhoven University of Technology (Fig. 2b) [23,24].

The idea of printing the building components separately in the factory erases the sophisticated process to connect walls and foundations. It provides a controlled curing environment for concrete [25] but takes more time and cost in transportation. Similar to typical prefabrication methods, preprinting projects include a series of operations such as relocation, storage, hoisting, and installation [26]. Besides, in preprinting projects, additional laboring steps are still required to reinforce or seal the gaps between assembled components. Although the deformation of printed concrete usually occurred within the first 12 h after extrusion, the compression and tensile strength of the material are unstable within the first 7 days [27]. If preprinted components are transported and assembled during this period, the above-mentioned process might affect their mechanical properties. If not, more time and space have to be counted in the preprinting process.

There are three common approaches to constructing the roofs of 3DCP houses. Affected by the esthetic considerations, environmental goals, local materials, and project budget, the choice of roofing types varies in different construction practices, but few procedures of them are truly realized by 3D printing. The first way is to build wooden or steel frames above 3D printed walls (Fig. 1c) [28] and then install roof panels on them (Fig. 1f, Fig. 3a) [29]. To adopt such a method, designers usually choose the shed roof to ensure that no additional process is needed to meet the drainage requirements. Also, the wood frame roof has lower environmental impacts, but it has slightly higher costs compared with the traditional reinforcement method [30]. The second approach is to assemble precast concrete slabs (Fig. 1d) and then print or install parapets (Fig. 1g, Fig. 3b) [31]. However, this approach has a great demand for formworks, especially when the plan of the house is irregular. That's because the formwork for irregular shapes has to be customized. The last one is to manually place formworks and bind rebars of the roof, then cast or spray concrete on them (Fig. 1e, Fig. 3c). Compared with the second approach, this method requires more on-site laboring procedures. Moreover, the process to assemble or print the parapet is still inevitable [32].

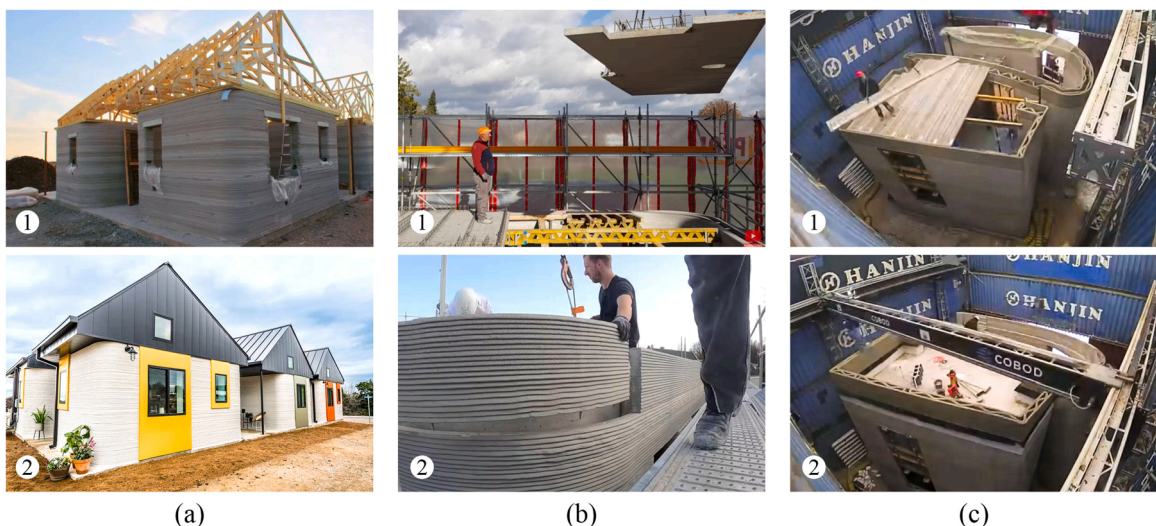


Fig. 3. (a) Construction records of the “Community First! Village - Homeless Housing” project from ICON Technology: (1) Wooden roof frames mounted above the 3D printed wall [28]; (2) Finished metal roof panels with gutters [29]. (b) Construction records of the Beckum House project from COBOD and PERI Group [31]: (1) Assembling the precast concrete slab; (2) Assembling the preprinted parapet component. (c) Construction records of the Kamp C project from COBOD [32]: (1) Placing formworks for concrete-casting; (2) 3D printing the parapets.

2.2. Problem statement

The above discussion indicates that the utilization of 3D printing in the real-scale building construction process is limited, and building a 3DCP house still requires lots of manual interventions. Firstly, neither the in-situ printing nor preprinting method can eliminate rebar-binding that exists in the procedures to connect walls and foundations or fill gaps between preprinted members. Secondly, since the mechanical properties of unreinforced concrete cannot tolerate the bending moments brought by horizontal structures, current 3DCP housing projects can barely realize 3D printed roofs. Only a few projects can print the parapets on-site or assemble preprinted members to ensure the integrated outlook of so-called “3D-printed” houses.

The limitations of 3D printing in realizing real-scale buildings not only come from the construction method and structural performance but also due to the mobility of the printers. Currently, neither the gantry printer nor the robotic arm can truly realize flexible and adjustable printing processes. On the one hand, projects adopting gantry systems have to print the building within the gantry framework, so designers of such projects tend to make full use of the printable space of the printer or choose a minimum size printer that could just contain the designed volume. Hence, there is no space left for additional printing works, like adding or replacing some components according to the adjustment of the design. On the other hand, in robotic arms projects, although robotic arms usually behave 6 degrees of freedom, the span of the arm is limited. Besides, the current in-situ robots cannot ensure the synchronization of movement and printing, they have to stop extruding materials when they are moving toward another position [21]. Therefore, manual procedures are still necessary to deal with the connection issue between different printed parts.

As discussed, three aspects that might cause additional manpower in architecture-scale 3DCP are summarized as follow:

- (1) The role played by 3DCP is limited in the building construction process, where many non-3D-printing procedures exist;
- (2) The current technology is unable to realize the 3D printing of all structural components, which makes it inevitable to use formworks and rebars to finish a building;
- (3) 3D printers lack flexibility, so the construction process should usually be fixed and can barely allow adjustment. Besides, some areas beyond the reach of printers have to be completed by manpower.

2.3. Objectives

Similarly, the Wujiashuang farmhouse project introduced in this paper aims to achieve the following goals (Table 1), so as to explore a design-to-printing workflow that is more suitable for 3D printing.

2.3.1. Reducing manual manufacturing procedures

The first goal of this study is to reduce manual manufacturing procedures in the process of 3D printing. Together, the development of the robotic 3D printing system (Section 3.2) and man-hour-saving construction details (Sections 4.2–4.4) help to realize the goal from technical and practical perspectives.

The evaluation criterion for the first goal is average man-hours, as elaborated in Eq. (1), this criterion is measured by man-hours cost per unit volume of the building entity (unit: h/cbm). In addition to the actual scenario of the project, this study will redesign the WJZ farmhouse by the other two construction methods, reinforced concrete (RC) structure and common 3DCP (only printing the walls), to compare their average man-hours with the proposed construction method (Appendix A). The average man-hours required for each work type are recalculated and summarized according to [33–35] and the specific recalculation equations are listed in Appendix B.

$$H = \frac{h_1 \times N_1 + h_2 \times N_2 + \dots + h_n \times N_n}{V_s} \quad (1)$$

Where H is the average man-hours to finish 1 cubic meter of the entire structure, h_n is the man-hours to finish a certain type of work (e. g. 3D printing, placing concrete, installing reinforcement, etc.), N_n is the number of workers required to finish a specific job, and V_s is the volume of the entire structure.

2.3.2. Realizing a completely 3D-printed house

The second goal of this study is to print the entire structure of the farmhouse, which requires comprehensive and systematic works

Table 1
Goals and tasks of the research.

Goals	Tasks to achieve the goals	Evaluation criteria
Reducing manual manufacturing procedures	Development and utilization of the mobile robotic 3D printing system	Average man-hours
Realizing a completely-3D-printed house	Man-hour-saving construction details Verifying the buildability of the house based on material design and structural evaluation	Participation of 3DCP works Mechanical tests and simulation
Improving construction flexibility	Construction site planning Organization of the printing process	Construction timeline

including the architecture design, material design, and structure evaluation (Section 3).

Two methods are adopted to evaluate the outcome of these works. Firstly, the feasibility of the printed structure is verified by the shaking table test and measured by the earthquake resistance according to China seismic intensity scale (CSIS). Secondly, the participation of 3DCP works in the entire farmhouse structure, quantified separately by volume and total man-hour, is calculated according to Eq. (2) and Eq. (3). Similarly, these quantification methods are applied in the common 3DCP design scenario, building a control group to check whether the design of a completely 3D-printed house can truly reduce the workload.

$$P_v = \frac{V_{3DCP}}{V_s} \quad (2)$$

$$P_t = \frac{T_{3DCP}}{T_s} \quad (3)$$

Where P_v is the proportion of 3DCP volume (V_{3DCP}) in the entire structure volume (V_s), P_t is the ratio of total man-hours consumed in 3DCP (T_{3DCP}) to the total man-hours consumed in the entire structure (T_s).

2.3.3. Improving construction flexibility

The third goal of this study is to improve the flexibility of 3D printing construction process. Namely, in-situ printing tasks are adjustable within each robotic arm, and various works are also adjustable among different robotic arms. In this study, the construction site planning (Section 3.3) and the organization of the printing process (Section 4.1) are the prerequisites to ensure a flexible use of the site.

The comparison of actual, ideal, and unplanned in-situ construction timeline is conducted to see whether the preplanning and organization of the construction process can enhance construction efficiency. To be specific, the actual timeline is summarized from the construction records of the Wujiazhuang farmhouse. By eliminating avoidable mistakes or relocating delayed works, an ideal timeline can also be rebuilt from the actual one. On the contrary, the unplanned timeline is an estimation assuming a linear workflow without any flexible adjustment and preplanning of the construction process.

3. Design strategies

3.1. Site context

The site context of the project is the prerequisite for the research team to start architectural design and printing process planning. The housing project introduced in this paper is located in a mountain village, Wujiazhuang, in northern Hebei Province, China (Fig. 4). The residents of the house are a peasant couple. The impacts of site conditions on the project are listed in Table 2, which can also be summarized into two aspects. The first impact comes from the limited construction area defined by the built environment, including the narrow alley and crowded surrounding houses. Therefore, it is more suitable for transporting robotic arms to the site for in-situ printing, which can not only utilize the 6-axis movement of robotic arms to reach the corners close to the neighbors' houses but also save the cost of assembling a large number of transportable but fragmented building components. The second impact comes from the unstable weather. The large temperature difference within a day and the dry environment of the mountain both require higher stability of materials in the printing process.

3.2. Mobile robotic 3D printing system

According to the space and weather challenges brought by the construction site, we develop a 3D printing system to improve the mobility of the robotic arm and stability of the material delivery (Fig. 5). In the system, the KUKA KR90-KR270 R2700 robotic arm is mounted on a mobile platform, including two track pairs and a lift table, to extend the mobility of the robot in X, Y, and Z directions. The previously fixed robotic base can now move freely on the site in these three directions (the movement on the X-direction tracks can reach 14 m, the Y-direction movement is about 1.3 m, and the Z-direction movement can increase by about 1.4 m). Additionally, a printing front-end is mounted on the 6th axis of the robotic arm, which expands the touchable radius of the robot by 1000 mm. With the above retrofit, three sets of robotic arms can cover the production area of about 16 m × 22 m on the construction site (Fig. 6). Specifically, Track B in the middle is a printing system with three-direction movement capability whereas Track A and Track C can already meet the required coverages by simply providing the X-direction movement.

The above-mentioned mobile platform also includes the transporting, mixing, and pumping system of the 3D printing material. The purpose of such a combination is to keep a constant relative position between the robotic arm and the material-delivery equipment when the 3D printing system is moving. By ensuring a short and fixed distance of the material transportation, it is possible to enhance the freshness of the extruded concrete and reduce the risk that concrete hardens in the delivery hose. Simultaneously, a spraying ring is mounted at the printing front-end to provide water vapor regularly during the printing process. This helps to avoid the drying shrinkage [37] of the concrete in its hydration process.

The controllers of the robotic arm, the print head, and the lift table are also integrated into the 3D printing system. Besides, the mobile platform provides a workspace for two technicians, which means that technicians don't have to move their positions in the printing process and their on-site workload can be effectively reduced.



Fig. 4. (a) The satellite image of the Wujiazhuang village; (b) The birds-eye view of the construction site; (c) The completed 3D printing house. (a) (source: Google Map); (b) (source: XWG Studio); (c) (source: XWG Studio).

Table 2

The impacts of site conditions on the project.

Site conditions	Impacts
Geography	The narrow roads in the mountainous area and the far distance from the urban area to the village both result in the high cost of transportation.
Weather	Large temperature differences within a day and the dry climate may affect the stability of the 3D printing concrete. Winter reaches earlier in the village than in the urban area, which shortens the desirable construction period.
Building code	Foundations shall not bear on frozen soil (depth: 1 m). Seismic precautionary intensity: VIII (China seismic intensity scale (CSIS)) [36].
Surrounding buildings	All construction equipment must be installed within the boundary line of the building. The east wall of the building shall be attached to the neighbor's house.

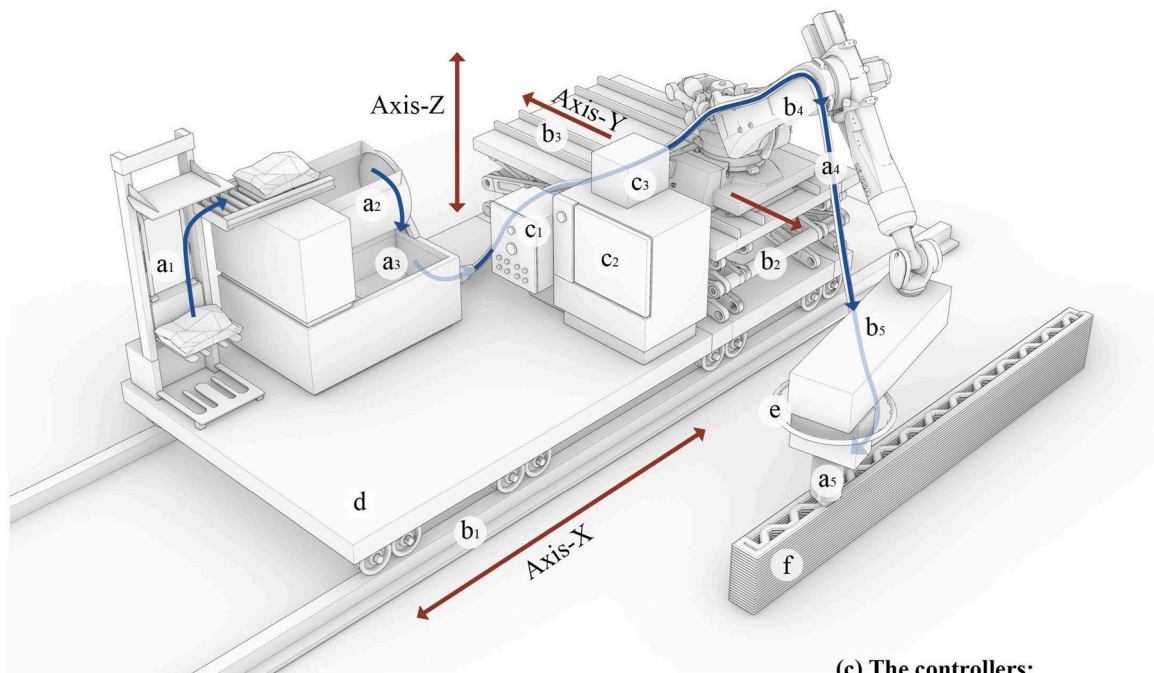
In this 3D printing system, the base of the robotic arm moves as shown by the red arrow in Fig. 5, and the material transportation process is marked by the blue arrow. To start the in-situ printing, the printing path of the robotic arms and the moving path of its base shall first be programmed in the KUKA|prc, a Grasshopper plugin. Then the program is imported into the robot controller, allowing the 3D printing system to move as planned on the construction site. Accompanying the transportation and extrusion of the concrete, the in-situ printing of this system can be realized.

3.3. Construction site planning

In the design of the farmhouse, we not only consider the requirements of daily use but also the technical ones of in-situ 3D printing. The construction site is about 24 m long and 22 m wide, covering an area of around 410 square meters. As discussed above, three sets of robotic printing systems are installed in parallel on the site (Fig. 6). In addition, all the 3D printed component types that make up the whole building are listed in Fig. 7, including the foundation trenches, structural walls, decorative walls, arch roofs, and flat roofs. It is worth mentioning that the anisotropic strength of the customized sulpho-aluminous cement has been tested beforehand (specimen size: $150 \times 150 \times 150$ mm, test machine: universal testing machine, loading rate: 0.5 MPa/s). The tested average compressive strength of the Z direction loading (Fig. 7) is 45.59 MPa (standard deviation $\sigma = 6.64$), and the average compressive strength of the X direction loading is 44.96 MPa ($\sigma = 3.44$). The result indicates that, even though the printing directions of these components are different, the anisotropic compression strengths from these directions are close enough to be treated as equal.

Fig. 6 also illustrates the location of the in-situ printed wall and the arrangement of the production area. The in-situ printed wall divides the farmhouse into three large bays and two small bays. Large bays are used as two bedrooms and a living room, they also define the in-situ printing zones of three robotic arms. Small bays are the in-between spaces of large bays, forming the bathroom and the kitchen of the house. Together, these in-situ printed walls enclose all the functional spaces of the farmhouse.

Besides the in-situ printed walls, the majority of the preprinting components are realized in the site preprinting zone. In other words, the track of each printing system is virtually divided into two parts. The north part is responsible for the in-situ printing that requires days of work for each wall whereas the south part is used for more flexible works that can be finished in just a few hours. In the actual printing process, some unpredictable adjustments may affect the pre-planned construction schedule, such as (1) modifying the design or print path, (2) path intersections between different printing systems, (3) maintenance of construction equipment, etc. And it is also necessary to spare enough time for unavoidable procedures like concrete hydration, printhead replacement, and infilling insulation materials. These processes may result in time intervals between different printing tasks. Therefore, the division of in-situ and preprinting zone can deal with such problems, if one of the tasks needs to be suspended in the above-mentioned situation, the robotic arm can move to another position of the track to start another task, which can improve the flexibility of the entire 3D printing process.

**(a) The material transportation system:**

- (a1) Material transporter;
- (a2) Concrete mixer;
- (a3) Concrete pump;
- (a4) PVC fiber reinforced delivery hose;
- (a5) Printhead.

(b) The path-operate system:

- (b1) X-direction tracks;
- (b2) Lift table;
- (b3) Y-direction tracks;
- (b4) Robotic arm;
- (b5) Printing front-end.

(c) The controllers:

- (c1) Lift table controller;
- (c2) Robot controller;
- (c3) Printhead controller.
- (d) Mobile platform.
- (e) Spraying ring.

Fig. 5. The developed mobile robotic 3D printing system.

3.4. Material design

The 3D printing material used in this project is fiber-reinforced concrete. The main components of the dry mix are sulpho-aluminous cement, sand, and polyvinyl alcohol (PVA) fibers. The specific formulations of the dry mix and the additives have already been presented in [38]. Different from the preprinting method in the factory, which provides higher quality control of the material [25], in in-situ printing, the workability of the 3D printing material can be easily affected by temperature and humidity changes. If the material ratio is not properly controlled, the fresh concrete will easily harden in advance during the delivery process, or the concrete cannot be solidified in time after being extruded from the nozzle. Therefore, how to enhance the environmental adaptability of the material is also a challenge for the project.

Considering that the water-cement ratio and the PVA fiber dosage are the determining factors for the mechanical property of the hardened concrete [38], to enhance the environmental adaptability of the printing material, we first ensure that the formulation of the dry mix remains unchanged, and then we adopt various additives to improve the workability of the fresh mortar.

Specifically, additives that control the material extrudability are the superplasticizer and the retarder, the former decelerates the slump loss of the fresh mortar [39] and the latter slows down the hydration process to minimize the changes in rheological properties of the concrete [40]. Increasing the dosage of these two additives can smooth the transportation process when the delivery speed of the concrete slows down or the material hardens earlier than usual. Also, additives that affect the material buildability are the accelerator and the viscosity modifier, the former helps the extruded concrete to obtain a short setting time to bear the stress from the upper layer [41] and the latter enhances the viscosity and cohesion of the material to improve the stability of the extruded shape [41]. Increasing the dosage of these additives will alleviate the situation that the deposited concrete cannot solidify in time and avoid the possible collapse of the printed component. Before the actual construction, we verified the feasibility of additives' dosage by workability tests (Table 3) and preprinting test (Fig. 8) in different temperature and humidity conditions. Accordingly, desirable additives dosages in various scenarios were summarized to ensure the stability of the material in the construction process.

Legend

Robot tracks:



(a) Robot's moving direction(s)

(b) Track bed

Production area:

- (1) In-situ printing zone
- (2) Site preprinting zone
- (3) Water supply
- (4) Garbage collection area
- (5) Material stacking area
- (6) Electricity supply

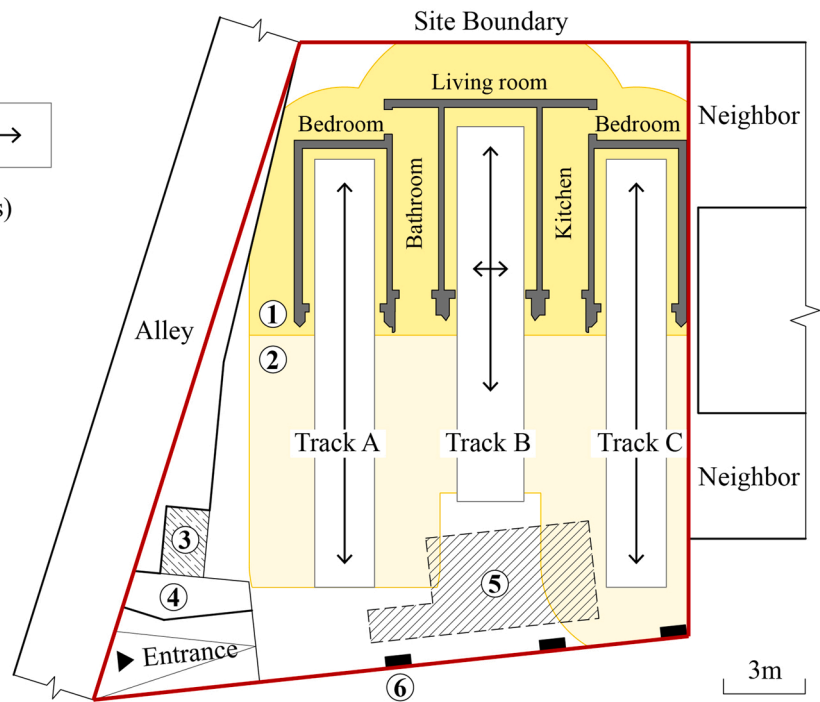
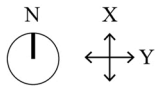


Fig. 6. The production area of the construction site.

Foundation Trench	Structural Wall	Decorative Wall	Arch Roof	Flat Roof
Printing Direction				
Horizontal	Horizontal	Horizontal	Vertical	Vertical
Loading Direction				
Printing Environment				
Factory preprinting	In-situ printing	Site preprinting	Site preprinting	Site preprinting

Fig. 7. The 3D printed component types of the farmhouse (Note: the vertical printing direction indicates that the printed component should be rotated by 90° in the assembling process).

3.5. Structure evaluation

A series of tests and simulations sounded the constructability of this farmhouse. This part of the work was assisted by Professor Peng Feng's Research Group at the Department of Civil Engineering, Tsinghua University, and the Architectural Design and Research Institute of Tsinghua University. Table 4 listed four specific structure evaluation methods. The first part was the concrete strength test

Table 3

Applicable workability parameters of the printing ink.

Factor		Requirements Maximum aggregates diameter (mm)		Test methods (Chinese standard)
		≤ 5	5–16	
Fluidity	Fluidity values (mm)	160–220	N/A	GB/T 2419 [43]
	Slump values (mm)	N/A	80–150	GB/T 50080 [44]
Curing time (min)		≤ 90		JGJ/T 70 [45]
Extrudability		Continuity, no clogging, no obvious crack in the pipe flow.		Observation
Buildability		The extruded shape remains stable and does not collapse.		Observation

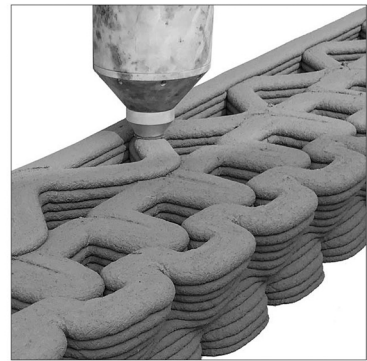
Source: Technical specification for concrete 3D printing (Chinese standard, T/CECS 786-2020) [42].



(a)



(b)



(c)

Fig. 8. (a) Test appropriate dosages of additives by gradually decreasing the extrudability of the material at a constant pumping speed; (b) Test appropriate dosages of additives by gradually reducing the buildability of the material in the same overhanging shape; (c) The 3D printing process of a decorative wall in the actual construction process.

(a) (Source: XWG Studio); (b) (Source: XWG Studio); (c) (Source: XWG Studio).

Table 4

Types and methods of the structure evaluation.

Evaluation types	Evaluation methods
Part 1: Concrete strength test	(1) A compression-testing machine is used to test the compressive strength of concrete cubes of the same size (compression area: 150 × 150 mm). (2) A hydraulic universal testing machine is adopted to test the axial tensile of concrete tensile specimens of the same size (cross-section size: 50 × 100 mm). (3) An electro-hydraulic servo universal testing machine is utilized to test the split tensile strength of concrete cubes of the same size (cross-section size: 150 × 150 mm).
Part 2: Finite element analysis (FEA) based on Midas Gen 2019	(1) To determine the wall thickness and the angle of the truss: Compare the mechanical properties of different design scenarios under wind loads, thermal actions, and seismic actions. (2) To choose the ideal truss shape of the arch roof: Compare the mechanical properties of different design scenarios under the permanent load, live loads, wind loads, thermal actions, and seismic actions. (3) To decide the hoisting method of the arch roof: Use the arch roof type selected in the previous step and compare the stress distribution of various hoisting scenarios in which the four lifting points are located at 1/4, 1/3, and close to 1/2 of the arch span (3 m). (4) To evaluate the buildability of the overall structure: Check whether the maximum stress of the overall structure (under the conditions of persistent design situations, transient design situations, and seismic design situations) meets the requirement of the concrete strength test.
Part 3: Mechanical test of the 3D printed wall	(1) Test method: Conduct quasi-static cyclic loading tests of different 3D printed wall samples. (2) Evaluation principles: Compare the mechanical performance of the column-strengthened wall with the traditional cast concrete wall.
Part 4: Shaking table test of the 3D printed scaled farmhouse model	(1) To make sure the safety of the entire 3D printed structure: conduct the shaking table test under different earthquake cases. (2) Evaluation principles: Code for Seismic Design of Buildings in China.

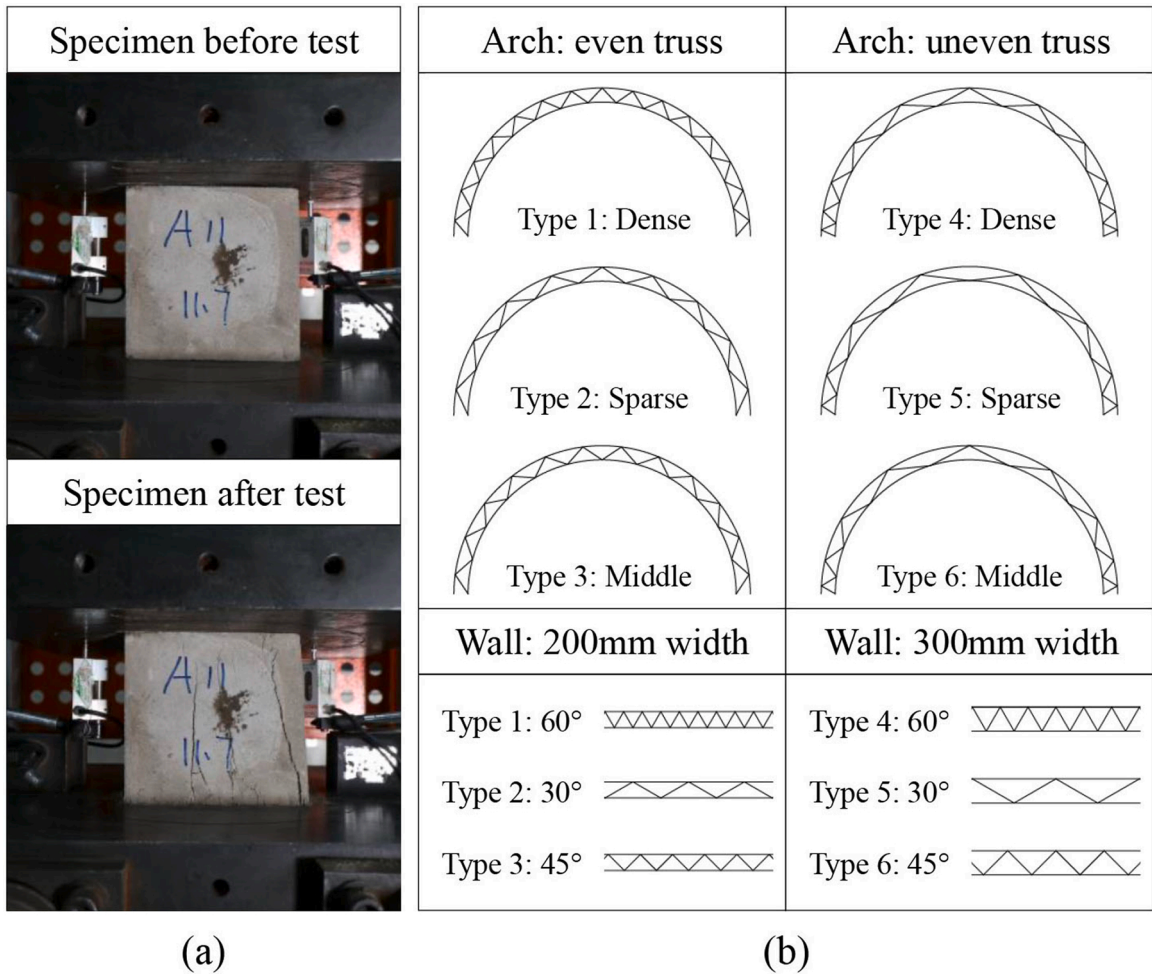


Fig. 9. (a) A specimen example of the concrete compressive strength test; (b) Different structural design scenarios of the arch roof and the load-bearing wall.

(a) (Source: XWG Studio).

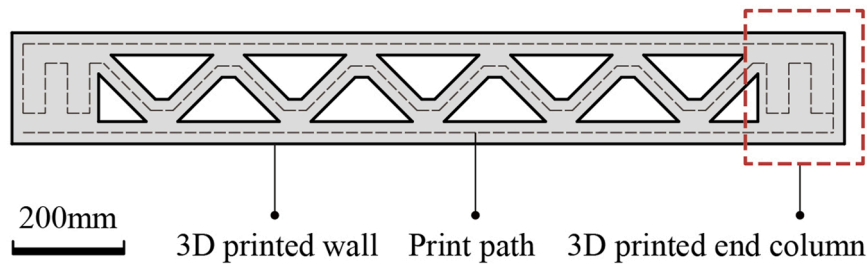


Fig. 10. An example of the column-strengthened wall (plan).

(Fig. 9a), the tested average compressive strength, axial tensile strength, and splitting tensile strength of the 3D printing concrete were 45.0 MPa (standard deviation $\sigma = 3.1$), 1.45 MPa ($\sigma = 0.15$) and 2.51 MPa ($\sigma = 0.25$), respectively. Based on the result of the strength test, the second part, finite element analysis (FEA), evaluated the mechanical properties of various structural design scenarios. Different types of the wall and the arch roof were analyzed in order to select the optimal structure for the farmhouse. Specifically, the load-bearing wall design scheme with the 45° truss (walls' type 3 and type 6 in Fig. 9b) was selected among six kinds of walls. At the same time, the arch roof design scheme with a thickness of 200 mm and even-spaced middle-density trusses (arch roofs' type 3 in Fig. 9b) was selected among six arch roof types. Compared with other design schemes, the maximum stresses of these selected types under different loading conditions were relatively small. A stress distribution simulation of the arch roof hoisting process also helped to decide that the most desirable position of the lifting points locate at 1/3 of the arch span. Then, the selected wall and arch roof were re-integrated into the building model to analyze the overall structure of the farmhouse. According to the analysis results, seismic design situations have the largest impact on the farmhouse structure, and the maximum stress among such situations, the tensile stress, met the requirement of the tensile strength test in Part 1.

Part 3 was the quasi-static cyclic loading test of the 3D printed wall with 3D printed end columns (Fig. 10). The safety of such a column-strengthened wall was ensured. Part 4 was the shaking table test of the scaled farmhouse model. The test proved that the farmhouse's structure performance met the goal of the seismic precaution specified in the Code for Seismic Design of Buildings (GB50011-2010) in China [36]: undamaged under small earthquakes, repairable under moderate earthquakes, and not collapse under large earthquakes.¹

4. Construction strategies

4.1. Organization of the printing process

The construction process of the Wujiazhuang farmhouse included four phases relating to foundations, walls, roofs, and finishing (Fig. 11). Phase 1 was the 3D printing and assembly of the foundation, which was conducted simultaneously at the construction site and the factory. On-site works included setting-out and site leveling, and the factory work was to print trench components of foundations. After these works were completed, the foundation trench components, together with the robotic arms, were transported to the site for assembly. Phase 2 was the in-situ printing of the structural walls. In this phase, decorative wall components were also printed at the site preprinting zone during the intervals between different in-situ printing works. Then, these preprinted components were hoisted and placed in their actual positions. In Phase 3, we continued the in-situ printing works and also printed the roof components in the site preprinting zone. Similar to Phase 2, all the roof components were hoisted and assembled at the end of this phase. The last phase is the surface treatment, the interior trim, and the landscaping of the open space. In this phase, the track beds of the 3D printing systems were kept and became three paths in the courtyard.

The above demonstration shows that 3D printing technology has involved the whole construction process of the farmhouse's structure. However, the difficulties of realizing such a process lie in how to assemble all building components in a convenient way, how to harness the flexibility of the robotic arm to deal with construction restrictions on the site, and how to truly reduce the labor force in the construction process. Correspondingly, the following four sections will introduce the key architectural details involved in each phase.

4.2. Foundations

The difficulties in foundation construction mainly lay in how to simplify the procedure of foundation reinforcements and foundation-to-wall connections. To solve these problems, we divided the foundation trenches into several components with an

¹ Small earthquake: the frequent earthquake action with Intensity being less than the local seismic precautionary Intensity. Moderate earthquake: the earthquake action with Intensity being equivalent to the local seismic precautionary Intensity. Large earthquake: rare earthquake action with Intensity being larger than the local seismic precautionary Intensity [36].

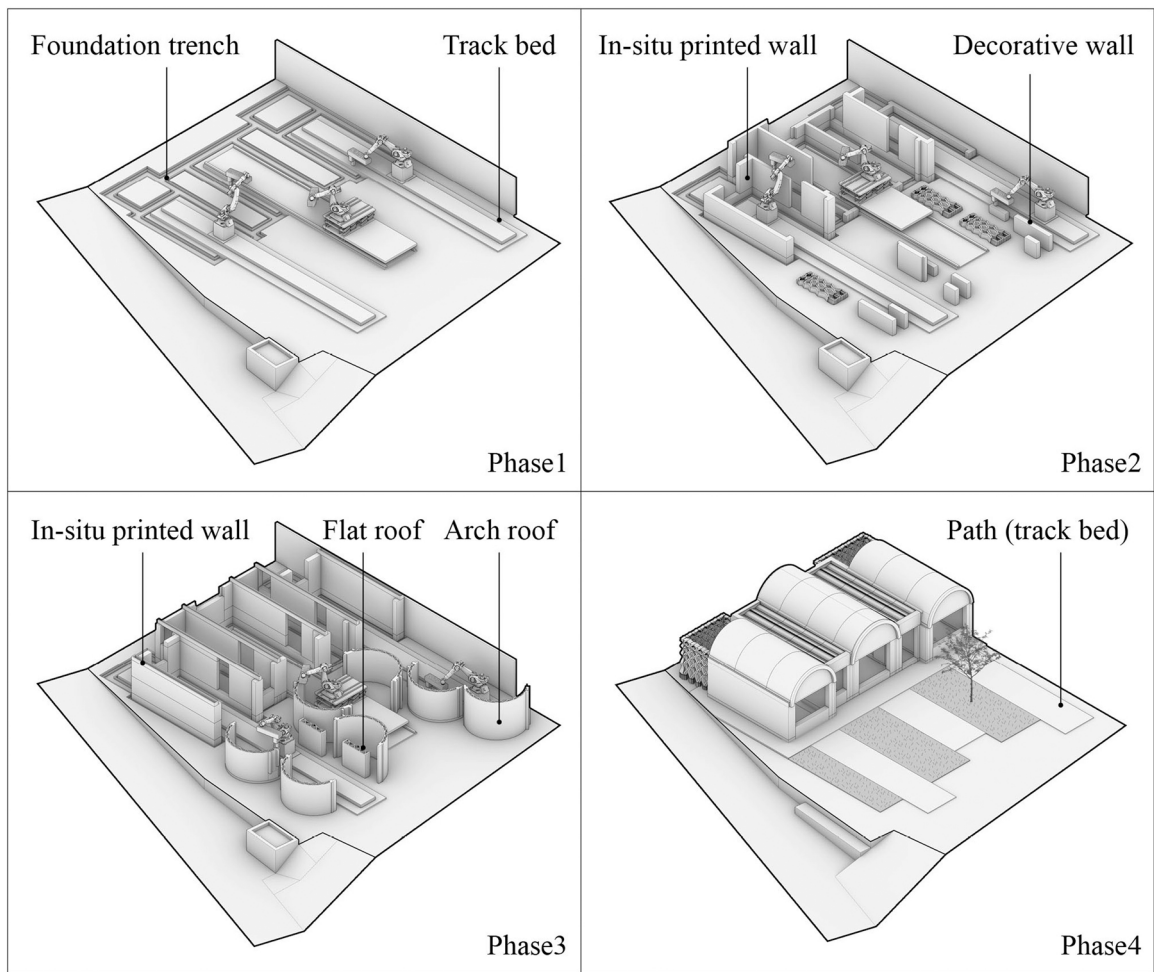


Fig. 11. The utilization of the site in each construction phase.

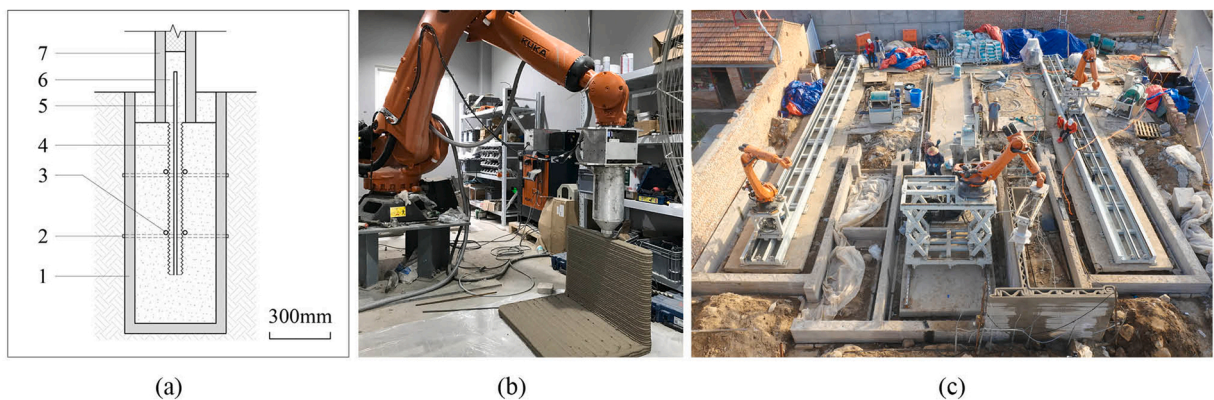


Fig. 12. (a) The connection details of foundations and walls (section); (a1) 3D printed concrete trench; (a2) X-directions FRP rebar; (a3) Y-direction FRP rebar; (a4) Flexible metallic conduit; (a5) Z-direction FRP rebar; (a6) Filled-in 3D printing mortar; (a7) 3D printed concrete wall; (b) The 3D printing process of the foundation trench; (c) The construction site after the walls' bottom were poured. (b) (Source: Shuyi Huang); (c) (Source: XWG Studio).

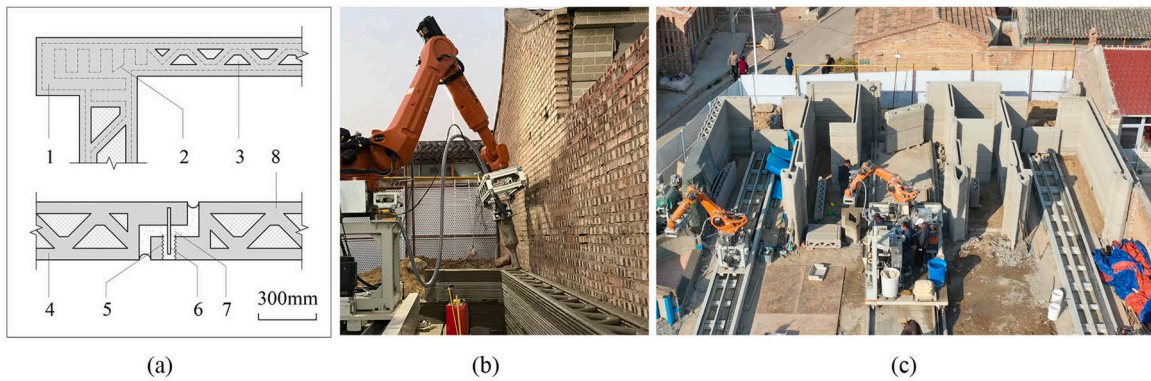
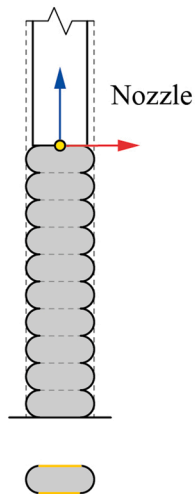


Fig. 13. (a) The construction details of the 3D printed wall (plan); (a1) In-situ printed wall; (a2) Print path to form the solid column; (a3) Spray polyurethane foam; (a4) In-situ printed wall; (a5) Flexible metallic conduit; (a6) FRP rebar; (a7) Filled-in 3D printing mortar; (a8) preprinted decorative wall; (b) The tilt-printing process; (c) The construction site in the in-situ printing process. (b) (Source: Shuyi Huang); (c) (Source: XWG Studio).

approximate length of 1 m for factory preprinting (Fig. 12b). During the printing process of each component, since the concrete already gained its buildability but had not completely solidified, fiber-reinforced plastic (FRP) rebars could be inserted in the X direction of the component (Fig. 12a). Therefore, in on-site assembly, workers only needed to tie the FRP rebars in the Y direction so that all horizontal rebars were connected. As mentioned in Section 3.5, there are trusses inside walls, if the Z-direction FRP rebars are directly tied to the trenches, that would inevitably lead to the collision between the printhead and rebars in the printing process. Hence, we adopted flexible metallic conduits (FMC) as the medium for connecting walls and foundations. Firstly, conduits were fixed with horizontal rebars, and then trenches were poured by 3D printing mortar up to the top of conduits, forming a flat surface to print the wall. The dry mix formulations of 3D printing mortar are similar to 3D printed concrete whereas such mortar has a higher water-to-cement ratio to

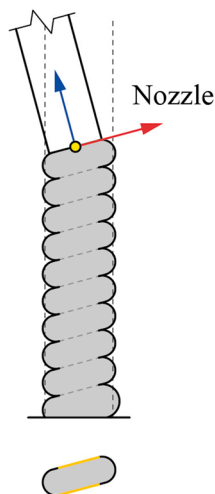
Print angle: 0°



Section area:
914.16 sqmm

Interlayer bond:
30 mm

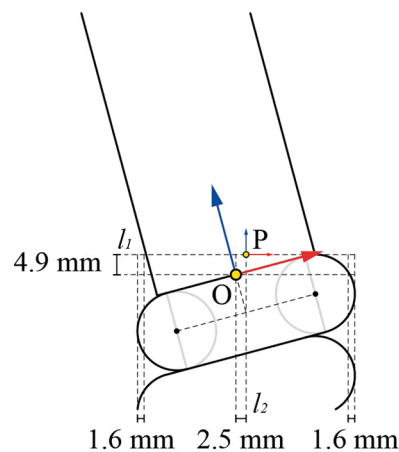
Print angle: 15°



Section area:
972.67 sqmm

Interlayer bond:
30 mm

Adjustment of the anchor point



P: original anchor point

O: new anchor point

l_1 : the highest elevation of extruded ink

l_2 : the axis of the printed concrete

Fig. 14. The adjustment of the nozzle's anchor point according to the 15° print angle.

allow better fluidity. Secondly, after the wall reached a certain height (around 300–400 mm), the Z-direction FRP rebars were dropped into conduits. Thirdly, we poured the bottom of the walls with 3D printing mortar, covering the Z-direction FRP rebars, to realize the connection between walls and foundations.

4.3. Walls

The key challenges in wall construction were the treatment of corners and building boundaries. According to the tests in Section 3.5, the feasibility of column-strengthened 3D printed walls had already been demonstrated. Therefore, the corners of all structural walls were printed as solid columns (Fig. 13a). The design of the print path allowed the integration of walls and corner columns, which saved the step of printing the mold of the column and then manually adding the reinforcement cage [8]. In the empty space inside the wall, we filled spray polyurethane foam (SPF) to improve the thermal insulation performance of the house. As for the connection between in-situ printed walls and preprinted decorative walls, on the one hand, we embedded FMC in the decorative walls, on the other hand, we inserted FRP rebars on each connection point on the in-situ printed wall. After a decorative wall was placed in its actual position, the FRP rebar could be inserted into the FMC (Fig. 13a). Then the space between the FMC and the rebar was filled with 3D printing mortar to realize the connection.

On the other hand, according to the local architectural customs and the limitation of the built environment, the wall on the east of the farmhouse should be attached to the neighbor's wall. To do so, the posture of the robotic arm was programmed to have a maximum tilt of 15°, hence the concrete can be deposited as close as possible to the neighbor's wall (Fig. 13b). To ensure the buildability of the concrete and the accuracy of the printed wall, the print speed and the position of the nozzle's anchor point were both adjusted according to Fig. 14. Firstly, to keep the minimum width of the interlayer bond as 30 mm, the section of the printed layer should be enlarged from 914.16 sq. mm to 972.67 sq. mm. Therefore, the printing speed was programmed to slow down by 6 %. Secondly, the location of the nozzle's anchor point was moved from point P to point O (Fig. 14) in order to keep the same top surface height of all printed walls. Eventually, a 5 cm gap was formed between the two walls and filled with SPF. It is worth mentioning that the tilt-printing method can only be realized by 6-axis robotic arms. If a gantry printer was adopted to do the job, its frame would have become an obstacle between the printhead and the existing wall, leaving a large space between the printed and existing walls.

4.4. Roofs

In the construction process of roofs, how to realize the connection between roofs and walls and how to meet the drainage requirement of flat roofs were the two crucial issues. Correspondingly, the “mortise-tenon joint” was adopted to connect the arch roof and the wall (Fig. 15a). On the one hand, trusses were erased in the last 20 cm of the wall's height, so this part of the wall remained hollow and became the “mortise” of the connection. On the other hand, shapes of “tenon” were printed on each side of the arch roof, so that the “tenon” could be embedded in the “mortise” of the wall. After arch roofs were all assembled, 3D printing mortars were poured into the gaps in-between the joint. As mentioned in Section 3.5, the optimal positions of arch roof lifting points had already been determined, hence these positions remained hollow in the printing process to form the holes for hoisting (Fig. 7), these holes were also re-filled with 3D printing mortar after assembly works were finished.

The flat roofs are the transitional structure between arch roofs, therefore, they became the accumulation area of the rainwater or snow water. Accordingly, the flat roofs were designed to have a slight slope to meet the requirements of drainage. Moreover, the wave

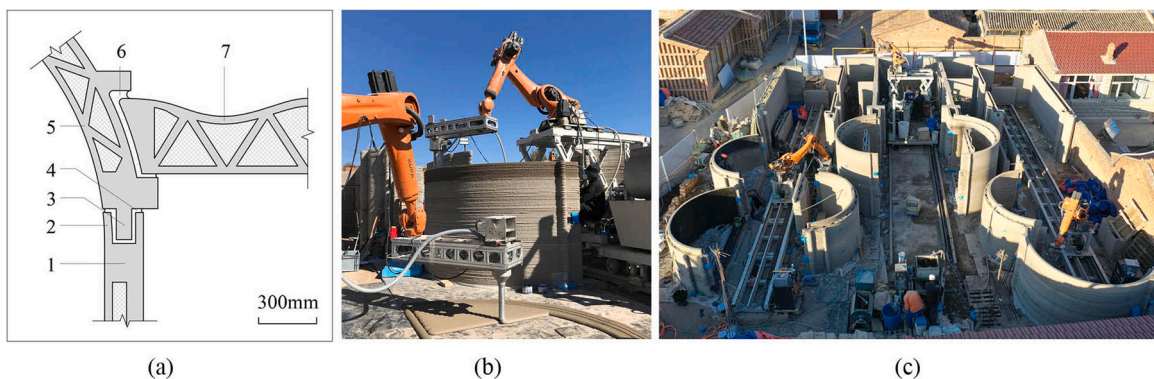


Fig. 15. (a) The connection details of walls, arch roofs, and flat roofs (section); (a1) In-situ printed wall; (a2) “mortise” of the wall-roof connection; (a3) “tenon” of the wall-roof connection; (a4) Filled-in 3D printing mortar; (a5) 3D printed arch roof; (a6) Filled-in 3D printing mortar; (a7) 3D printed flat roof; (b) The printing process of arch roofs; (c) The construction site after the arch roofs were printed.

(b) (Source: Shuyi Huang); (c) (Source: XWG studio).

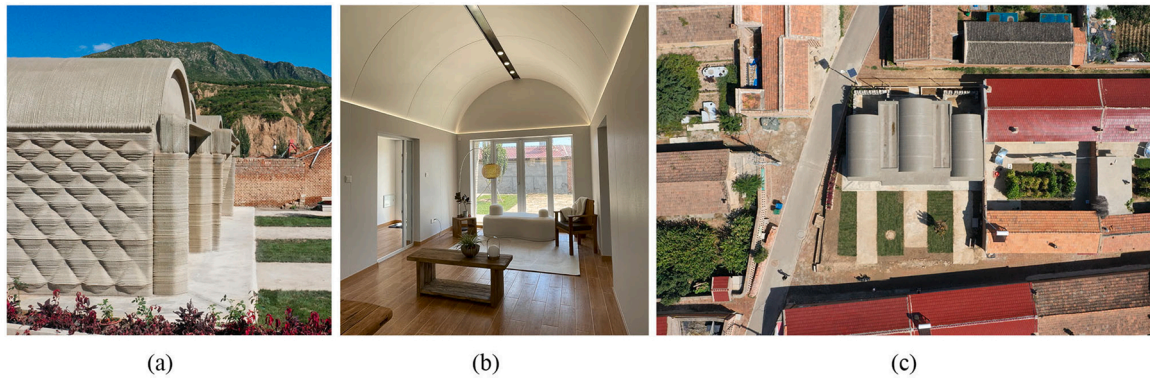


Fig. 16. The finished 3D printed farmhouse: (a) West perspective; (b) Living room; (c) Birds-eye view. (Source: XWG studio).

Table 5
Evaluation criteria and control groups of the actual construction scenario.

Evaluation criteria	Control groups
Average man-hours	RC structure design scenario Common 3DCP design scenario
Participation of 3DCP works	Common 3DCP design scenario
Construction timeline	Ideal construction timeline Unplanned construction timeline

Table 6
Man-hours consumption calculation of the WJZ farmhouse.

Structural component	Type of work	Man-hours (h)	Component volume (cbm)	Man-hours per unit (h/cbm)
Foundations (Phase 1)	Preprinting and on-site installation	648	44.01	14.72
Walls (Phase 2)	In-situ printing and casting	516	44.09	11.70
Roofs (Phase 3)	In-situ printing and insulation in-filling	604	44.09	13.70
	Site preprinting and installation			
	In-situ printing			
	Site preprinting and installation (including backyard decorative walls)			
Total man-hours (h)				1768
Total volume (cbm)				132.19
Average man-hours (h/cbm)				13.37

shape of the flat roof also formed a shallow trench to guide the water, preventing the water from infiltrating into the gap between the flat roof and arch roof (Fig. 15a). Additionally, the flat roof component located at the south or north end of the farmhouse was also printed together with the parapet (Fig. 7), which made the construction process of parapets no longer an additional procedure in the 3D printing project. Benefiting from the above design, the construction process of roofs was completely realized on the site, which solved the problem that the roofs need to be treated separately in conventional 3DCP building projects and reduced the reliance on manpower, transportation, and formworks in the construction workflow.

4.5. Finishing

After the printing and assembly process, three main procedures transformed the 3D printed concrete structure into a livable house (Fig. 16). The first procedure is the treatment of the exterior surface, including the mortar-sealing of gaps between components, the waterproof painting on the roof, and the installation of doors and windows. The second step is the interior trim. Fixed wood panels

Table 7

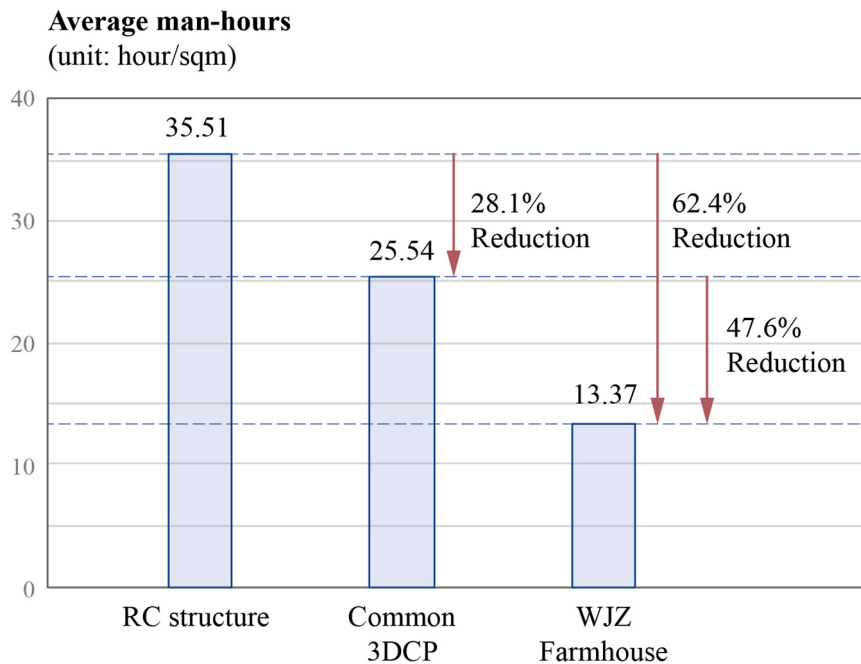
Man-hours consumption of the RC structure design scenario.

Work type	Volume (cbm)	Productivity (h/cbm)	Total man-hours (h)
Regular concrete components	55.11	22	1212.42
Irregular concrete components	47.62	59	2809.58
Reinforced concrete beams	4.04	26	105.04
Reinforced concrete columns	2.77	31	85.87
Block walls	18.43	18	331.74
Total	127.97	N/A	4544.65
Average man-hours (h/cbm)			35.51

Table 8

Man-hours consumption of the common 3DCP design scenario.

Work type	Volume (cbm)	Productivity (h/cbm)	Total man-hours (h)
Regular concrete components	47.52	22	1045.44
3DCP components	51.86	12	622.32
Irregular concrete components	29.47	55	1620.85
Reinforced concrete beams	4.04	26	105.04
Total	132.89	N/A	3393.65
Average man-hours (h/cbm)			25.54

**Fig. 17.** Average man-hours of different design scenarios.

were installed on the surface of the 3D printed interior wall through steel keels, which increased the functionality of the interior space (Fig. 16b). The last procedure is to reorganize the open space in front of the farmhouse. As mentioned before, track beds of the printing system were kept as the paths in the courtyard. Moreover, three lawns were enclosed by these paths as the peasants' recreational and planting space (Fig. 16c).

5. Evaluation

Three evaluation criteria proposed in Section 2.3 are calculated based on the actual construction scenario of the Wujiazhuang farmhouse. As mentioned before, different construction methods and workflows will be adopted as control groups (Table 5) to demonstrate the performance of the proposed design-to-printing workflow.

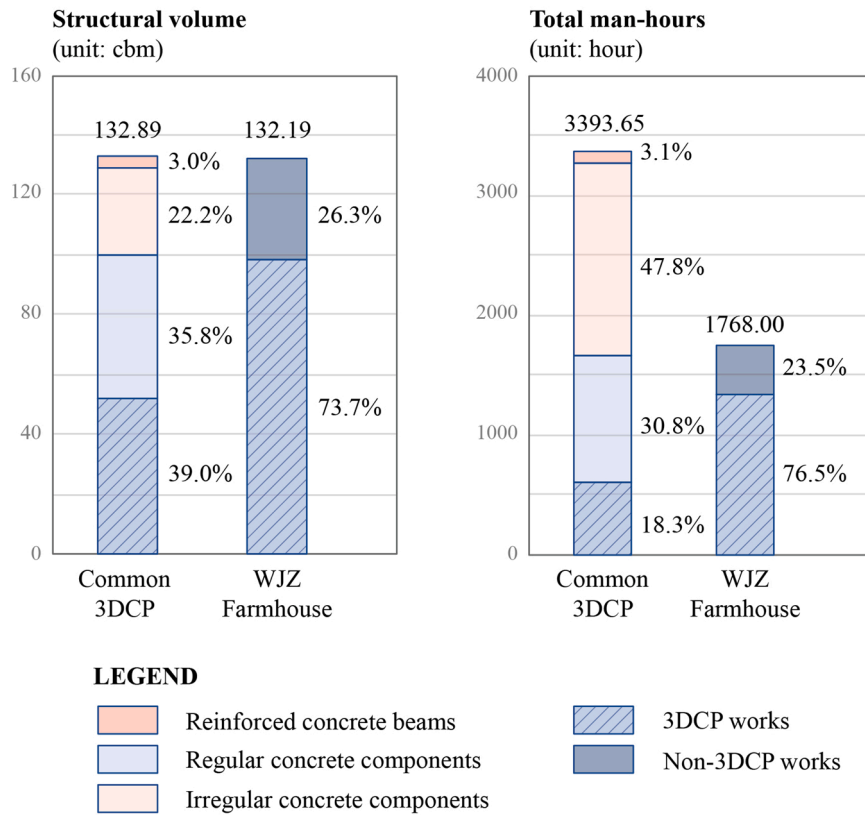


Fig. 18. Participation of 3DCP works measured by structural volume and total man-hours.

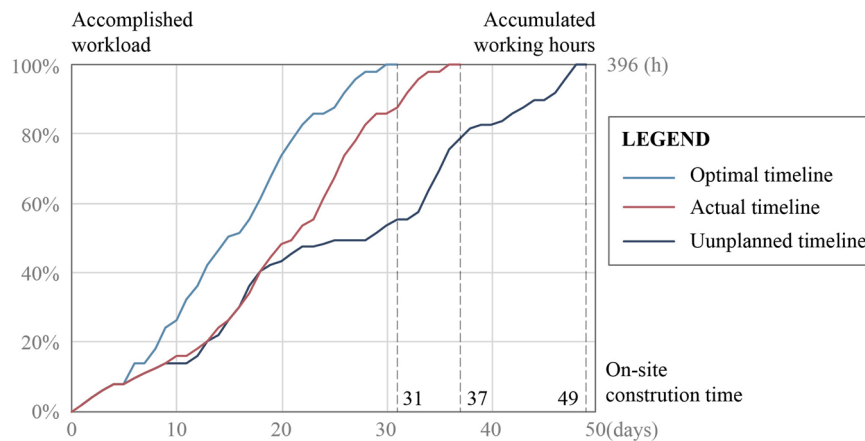


Fig. 19. Construction timeline of different construction scenarios.

5.1. Average man-hours

The average man-hours consumption of the Wujiazhuang farmhouse structure is 13.37 h/cbm, which is the ratio of the total man-hours to total structural volume (Table 6). Concretely, the total man-hours is the accumulation of the total working time of all workers in the construction process, and the total structural volume excludes the volume of window, doors, and interior trim. The maximum number of daily workers number is 8 people, and the maximum daily working time is 8 h. Differently, according to Eq. (1) and Appendix B, average man-hours of detailed work types are summarized in Tables 7 and 8 to estimate the average man-hours of the other two design scenarios. The result indicates that the RC design scenario will consume 35.51 h/cbm of average man-hours and the common 3DCP method will cost 25.54 h/cbm. That means, compared with the traditional RC construction method, the common 3DCP

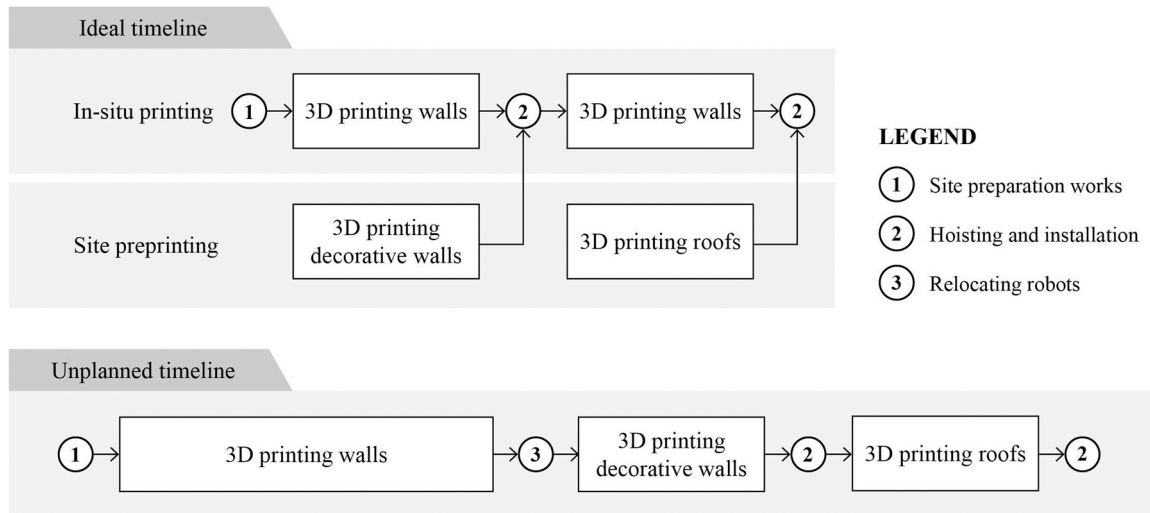


Fig. 20. The concept of estimating the ideal and unplanned workflow.

approach can only reduce human resources by 28.1 % while the proposed design-to-printing workflow in this research can realize a 62.4 % reduction (Fig. 17).

5.2. Participation of 3DCP works

As listed in Table 6 and Table 8, the total man-hours of the common 3DCP scenario are 3393.65 h whereas the actual design scenario is only 1768 h even if their structure volumes are quite close (note: the difference in construction method results in a slight difference in structure volume). That's because the participation of 3D printing technology in WJZ farmhouse exceeds the one in common 3DCP option (Fig. 18). Particularly, only 39.0 % of the entire structure is realized by 3DCP in the control group, which only corresponds to 18.3 % of the total man-hours. That is, only printing the walls of the house will still consume a lot of human resources. In contrast, the 3D-printed part of the actual design accounts for 73.7 % of the whole structure, which is equivalent to 76.5 % of total man-hours, significantly higher than the figure of the typical 3DCP choice. Namely, printing the whole structure of the building is a more manpower-saving approach. It is worth mentioning that, according to the local building code, the depth of the farmhouse's foundation needs to reach 1 m, so the volume of the placed 3D printing mortar is remarkably large, accounting for 96.5 % of the volume of non-3D printing part. Besides that, only the side panels of the arch-shape roofs and the gaps between preprinting components are filled by 3D printing mortar. In other words, this building is a completely 3D-printed house.

5.3. Construction timeline

The outcome of construction site planning and organization is evaluated by the comparison of the ideal timeline (31 days), actual timeline (37 days), and unplanned timeline (49 days). Fig. 19 records the construction timeline starting from the first day of in-situ 3D printing and ending with the day when all components are correctly located. The ordinate indicates the accomplished workload that is measured by daily 3D printing hours on the construction site. Precisely, in the unplanned timeline, we assume a construction scenario without the definition of the in-situ printing zone and preprinting zone (Fig. 20), which means all the tasks have to be finished step by step, forming a linear workflow. In contrast, the ideal timeline demonstrates an optimal scenario maximizing daily workload starting from the first day. In fact, due to the influence of the outdoor environment, a couple of days were spent debugging the equipment and adjusting material formulations in the actual project, and the synchronous printing of three robotic arms could not be realized immediately, which led to the difference between the ideal timeline and the actual timeline. In general, by increasing the flexibility of the construction site, the in-situ construction duration can be reduced by 24.5 % compared with the unplanned choice.

6. Discussion

6.1. Paradigm shifts in concrete construction techniques

From the perspective of automated construction, the research explored the possible paradigm shifts in digital concrete construction through the following aspects:

Firstly, the robotic 3D concrete printing system proposed in this paper makes it possible to organize the construction process more flexibly. The utilization of the 3DCP technology in the construction site should not be limited to letting one printer work continuously, in contrast, designers should try to make full and diverse use of multiple devices in various tasks. The advantage of the introduced robotic 3D printing system is that it can print different building components in the site preprinting zone several times while accomplishing the task of in-situ printing. That is to say, the printing process is not a one-off and error-free one, but a more flexible one that allows real-time adjustment according to the construction requirements. On the contrary, the existence of the girder in the gantry printer prevents a project from do-over tasks since the girder will collide with the printed wall if it tries to move back to the ground. Therefore, the potential of robotic 3D printing is more promising considering the goal to realize flexible construction.

Secondly, the design-to-printing workflow of the farmhouse confirms that it is practical to 3D print the entire structure of a building. Although building components are not necessarily made of the same material, for the burgeoning 3D printing technology, the attempt to 3D print all building components has two aspects of significance. On the one hand, the dependence on the formwork in concrete construction is further reduced. Once precast or cast-in-place components are involved in the construction process, the large demand for formworks is inevitable, especially in projects with irregular forms. That's because irregular structures usually require expensive, customized formwork, which can barely be reused or even recycled [46]. On the other hand, the whole-process 3D printing explores the possibility of weaving the form and function of each type of building component as the material can be positioned at the exact point where they are needed [47]. For example, using the 3DCP technology, designers can combine the slope and the wave-shaped trench in the flat roof or create parametric texture for the wall without additional construction expense, which, on the contrary, may require more in-process materials or manpower if they choose the traditional construction method. In short, the 3D printing of all building components is not only an attempt to realize the mold-less shaping of concrete [12] but also provides a better balance between the form and function of structural components.

Finally, the entire project explores the potential of harnessing the 3DCP technology to solve construction problems and improves the participation of such a technology in the construction process. As it is claimed in [47], rather than operating within the predefined programs, we should design the actual "manual skill" of the robots. Therefore, when problems popped up in the construction process, we tried to figure out the strategies requiring no extra labor force. For example, the connection issue of the walls and foundations was solved by the pre-inserted FMC, and the requirement to attach the wall to the neighbor's house was met by the tilt-printing method. Furthermore, the introduction of numerical control technology, coupled with the 6-axis movement of the robotic arm, had technically transformed the construction site into a digital space. Therefore, once the position of the robotic arm was settled, many jobs on the construction site, including measuring, positioning, and leveling, were assisted by the robot, which not only reduced the demand for manpower in these jobs but also improved the safety of the site. That is to say, the role of the robotic 3DCP technology is no longer limited to 3D printing but comprehensive participation in the construction process.

6.2. Future studies

The design-to-printing process of the farmhouse also reveals several topics that are worthy of further research:

The first topic is the adaptability of compression-only structures. With small bending moments, the arch roof adopted in this project can meet the span requirement of interior living space. But that kind of compressed-loaded application seems to be the only solution for current 3D printed roofs [46]. The technology cannot realize the printing of large-span flat roofs or slabs because the bending moments of these structures are non-negligible. Although the 3D printed concrete beam [48] and the lightweight composite concrete slab [49] have been researched in recent years, the real-scale horizontal structure that can truly be realized by mold-less shaping has not yet appeared. The key approach to solving this problem is the utilization of new materials. Compared with the commonly used Portland cement or sulpho-aluminous cement, the Engineered Cementitious Composite (ECC) and the Ultra-High Performance Concrete (UHPC) are confirmed to have better compressive and tensile properties [50,51], however, these materials are not printable currently. Therefore, how to develop materials with better mechanical properties as well as printability, and how to upgrade the printers to allow the 3D printing of more material types are possible research directions in the field of 3DCP.

The climate adaptability and the working duration of in-situ printing are also worth further studying. In the project, even though several preprint tests had been carried out before the actual printing, the material transportation process was still unstable occasionally. That's because the temperature and humidity in mountain areas varied greatly, making it hard to simulate all the possible weather conditions beforehand. Besides, as the printing time went longer, the possibility that materials adhered to the delivery system (including the pump, the delivery hose, or the printhead) went up, which affected the efficiency of material delivery. Such a problem usually occurred after continuous printing for more than 6 h. Without immediate treatment, the accumulation of hardened concrete would cause blockage in the delivery system and the printing process had to be suspended or terminated. Therefore, how to ensure the stability of material delivery and realize long-duration printing under unstable weather conditions are topics requiring investigation.

7. Conclusion

The completion of the Wujia Zhuang farmhouse demonstrates that the adopted design-to-printing workflow is practical to increase the use of automation on the construction site. As summarized in Fig. 21, tasks included in such a workflow effectually contribute to realizing the proposed goals in Section 2.3.

Whether the task helps to achieve the goal		Achieved goals	62.3% Man-hours saving	76.5% 3DCP works participation	24.5% On-site duration saving
Solved tasks					
Design-to-printing workflow	Mobile robotic 3D printing system	✓	✓	✓	
	Construction site planning				✓
	Material design			✓	
	Structural design and evaluation			✓	
	Organization of the printing process				✓
	Man-hour-saving construction methods	✓	✓		

Fig. 21. Contributions of design-to-printing workflow to automated construction.

Compared to common 3DCP research and projects, the innovations and advantages of this paper can be concluded as follow:

- ◆ The development of the mobile robotic 3D printing system provides a novel printing method that not only allows 6-axis movement but also ensures simultaneous operations of moving and printing. The idea of integrating the robotic arm, material mix-and-delivery system, and mobile platform is applicable in other in-situ 3D printing projects to realize architecture-scale construction.
- ◆ As a 3D printing material with better environmental performance, the sulpho-aluminous cement is verified to be buildable in the completely 3D-printed house. The proposed structure design and evaluation methods to utilize such a particular material in 3DCP construction is proved to be reasonable and practical because of the realization of the farmhouse.
- ◆ Rooting on a unique printing scenario with compact built environment, the research not only demonstrates how to balance the design and production requirements in a limited and isolated space but also introduces empirical construction methods to simplify the manual procedures, increase construction efficiency, and maximize the use of robots.
- ◆ The proposed design-to-printing workflow realizes a printing process that saves both human resources and time. Accordingly, the tasks, evaluation criteria, and numerical results presented in this paper are significant add-ups to the current research on large-scale 3DCP.

All in all, the introduced workflow is the first step from 3D printing an entire one-story house to the truly automated constructions of various buildings. It also provides a new perspective, pointing toward a parametric, intelligent, and flexible future in the construction industry.

Funding

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

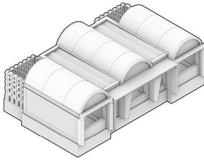
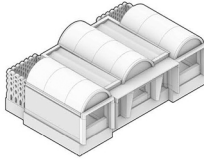
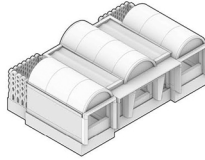
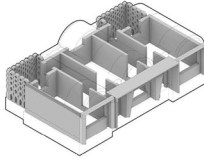
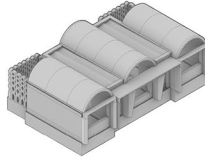
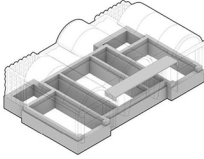
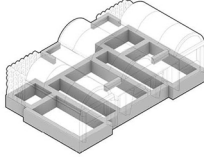
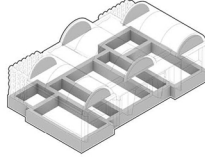
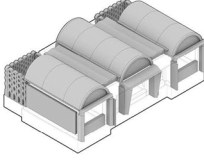
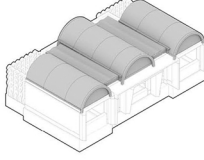
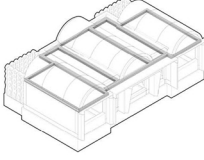
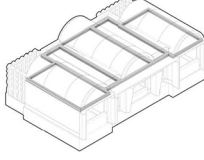
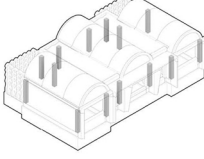
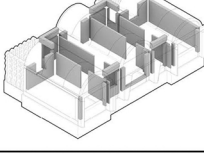
Data Availability

The data that has been used is confidential.

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Appendix A. Digital models of different design scenarios to construct the Wujiazhuang farmhouse

Reinforced concrete structure	Common 3DCP Only print the walls	Actual design Completely 3DCP
Structure entity		
		
3DCP components		
N/A		
Regular concrete components		
		
Irregular concrete components		
		N/A
Reinforced concrete beams		
		N/A
Reinforced concrete columns		
	N/A	N/A
Block walls		
	N/A	N/A

Appendix B. Data references to estimate man-hours

Ref.	M. Batikha [33]	B. de Soto [34]	Y. Weng [35]
Structure type	2-story building	Straight wall	Curve wall
Construction method	RC	RC	RC
Measurement of productivity	Man-hours per unit area	No. of workers, time	Total man-hours
Unit	hour/sqm	hour	hour
Transformation equation	$h = h_a \times A/V_s$	$h = N \times T_w/V_s$	$h = T_{sum}/V_s$
Referred data			
Foundation/slab/wall	N/A	19.97	59.00
Block wall	18.41	N/A	23.94
Beam	26.28	N/A	21.43
Column	31.10	N/A	N/A
Mortar placing	N/A	N/A	N/A
3DCP	N/A	5.72	14.30
		N/A	4.79
			N/A
			12.12

Note:

h: average man-hours per unit volume to finish a certain type of work (h/cbm).

h_a : average man-hours per unit area to finish a certain type of work (h/sqm).

A: building area.

V_s : volume of the structural component.

N: number of workers.

T_w : working hours of each worker.

T_{sum} : total working hours of all workers.

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